

INCIDENT REPORT

IR-2026-001

Classification	CONFIDENTIAL	Severity	HIGH
Incident ID	IR-2026-001	Status	RESOLVED
Date	May 21, 2026	Analyst	Ronak Mishra

1. Executive Summary

A brute force SSH attack was detected against the Ubuntu server (ronak, 10.0.0.33) originating from an internal Kali Linux machine (10.0.0.100). The attack consisted of 260 failed SSH authentication attempts across three distinct waves spanning approximately 28 minutes. The attack was detected automatically by Microsoft Sentinel using a custom KQL scheduled analytics rule ingesting Linux Syslog data. No successful authentication was achieved at any point during the attack. The attacker IP was contained via a UFW deny rule on the Ubuntu endpoint within 17 minutes of detection.

2. Incident Timeline

Time (UTC)	Event
05:05	First wave of SSH brute force begins — 48 failed authentication attempts
05:08	Microsoft Sentinel analytics rule detects authentication failure spike
05:20	Attack pauses briefly — attacker tool reloading wordlist
05:25	Second wave begins — 120 attempts in under 5 minutes (highest intensity)
05:30	Third wave — 88 additional attempts
05:33	Attack activity stops — 260 total failed attempts logged
05:45	Analyst identifies attacker IP (10.0.0.100) and targeted account via KQL forensic investigation

05:50	Containment actioned — UFW deny rule deployed: sudo ufw deny from 10.0.0.100 to any port 22
05:52	Containment confirmed — SSH connection from Kali Linux timed out, no further access possible

3. Attack Details

Field	Value
Attack Type	SSH Brute Force (Credential Access)
Attack Tool	Hydra v9.x
Attacker IP	10.0.0.100 (Kali Linux 2025.2 VM)
Target Host	ronak (Ubuntu 22.04, IP: 10.0.0.33)
Target Service	SSH (Port 22)
Target Username	ronak
Total Attempts	260 failed authentication attempts
Attack Duration	~28 minutes (05:05 – 05:33 UTC)
Successful Auth	None — attack failed
MITRE Tactic	TA0006 — Credential Access
MITRE Technique	T1110.001 — Brute Force: Password Guessing

4. Detection Method

The attack was detected automatically by Microsoft Sentinel using a custom scheduled analytics rule. The rule queries Linux Syslog data ingested from the Ubuntu endpoint via Azure Monitor Agent. It fires when more than 10 failed SSH authentication attempts are detected within a 5-minute window from the same source, which exceeds normal operational thresholds and is indicative of automated brute force activity.

KQL Detection Query:

```
Syslog
| where Facility == "authpriv"
| where SyslogMessage contains "Failed password"
| or SyslogMessage contains "authentication failure"
```

```
| where Computer == "ronak"  
| summarize FailedAttempts = count() by Computer, HostName, bin(TimeGenerated, 5m)  
| where FailedAttempts > 10
```

Rule Configuration:

- Severity: High
- Schedule: Runs every 5 minutes, lookback window 5 minutes
- Threshold: More than 10 failed attempts per 5-minute window
- MITRE ATT&CK: T1110.001 — Brute Force: Password Guessing
- Incident creation: Enabled — auto-generates incident on trigger

5. Investigation Findings

KQL forensic investigation confirmed the following findings:

- 260 failed SSH authentication attempts originated from 10.0.0.100 between 05:05 and 05:33 UTC
- All attempts targeted a single username: ronak
- Attack occurred in three distinct waves: 48 attempts, 120 attempts, 88 attempts
- No successful authentication was recorded at any point — EventID pattern shows only failed logins
- No lateral movement detected — attacker did not pivot to any other system
- No data exfiltration detected — no outbound data transfer observed
- No persistence mechanisms identified — no new accounts, cron jobs, or SSH keys created
- System integrity maintained — no files modified, no services altered
- Ubuntu firewall (UFW) was active but SSH port was open prior to containment

6. Containment Actions

1. Opened Microsoft Sentinel incident (ID 1) and assigned to analyst Ronak Mishra — status set to In Progress
2. Ran KQL forensic investigation to confirm attacker IP (10.0.0.100), targeted account, and attack timeline
3. Verified no successful authentication occurred by checking for absence of successful login events
4. Deployed UFW deny rule on Ubuntu endpoint to block attacker IP on port 22:

```
sudo ufw deny from 10.0.0.100 to any port 22  
sudo ufw status      # Confirmed rule active
```

5. Verified containment — attempted SSH connection from Kali Linux (10.0.0.100) timed out with no response
6. Confirmed analytics rule remains active for ongoing monitoring of future attempts

7. Incident status updated to Resolved in Microsoft Sentinel

7. Recommendations

#	Recommendation	Priority
1	Implement SSH key-based authentication and disable password authentication entirely in <code>/etc/ssh/sshd_config</code>	Critical
2	Deploy fail2ban to automatically block source IPs after a configurable number of failed authentication attempts	High
3	Restrict SSH access to known IP ranges or VPN-only access via UFW allowlist	High
4	Configure Microsoft Sentinel alert to notify analyst immediately when SSH brute force incident is created, reducing response time	Medium
5	Consider relocating SSH service to a non-standard port (e.g. 2222) to reduce automated scan exposure	Low

8. Lessons Learned

- Custom KQL analytics rules in Microsoft Sentinel successfully detected the brute force pattern within 3 minutes of the first failed attempt, demonstrating the value of fine-tuned detection thresholds over relying on default rules.
- The multi-endpoint architecture with Azure Arc and Azure Monitor Agent enabled centralized log correlation across both Windows and Linux endpoints from a single Sentinel workspace — critical for enterprise-scale monitoring.
- Rapid containment via UFW was effective but manual. Automating this response via the SOAR playbook (configured in Day 6) would reduce mean time to respond (MTTR) from 17 minutes to under 1 minute.
- Password-based SSH authentication presents an unacceptable risk surface. Key-based authentication should be enforced as a baseline security control on all Linux endpoints.
- The absence of fail2ban allowed all 260 attempts to reach the SSH daemon. Progressive blocking would have cut the attack short after the first 5-10 attempts.

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